

MAUN, Botswana

WMO: 680320

Lat: 19.973S

Lon: 23.431E

Elev: 3093

StdP: 13.13

Time Zone: 2.00 (E02)

Period: 97-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	44.5	47.4	15.8	13.7	64.8	20.4	17.2	65.7	18.6	65.5	16.7	66.6	5.2	140	0.373

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
10	26.7	100.6	65.8	98.8	65.8	97.0	66.1	74.5	85.7	73.7	85.1	73.0	84.3	9.0	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
71.9	132.3	77.1	71.3	129.9	76.8	70.2	124.6	76.3	40.4	85.9	39.6	85.5	38.9	84.4	79.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
18.3	16.2	14.3	DB	39.3	104.6	3.1	2.8	37.1	106.5	35.3	108.2	33.5	109.7	31.3	111.7
			WB	32.3	77.2	2.9	1.5	30.2	78.3	28.5	79.2	26.9	80.0	24.9	81.0

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	75.2	80.0	79.9	78.2	75.2	69.3	64.6	63.7	69.5	77.3	82.8	81.6	80.5	(6)
	DBStd	7.63	3.82	2.95	3.81	3.59	4.23	3.82	3.64	4.89	4.64	4.25	4.92	3.79	(7)
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0	(8)
	HDD65	149	0	0	0	1	11	52	70	15	0	0	0	0	(9)
	CDD50	9195	930	838	874	756	600	438	424	604	820	1016	948	947	(10)
	CDD65	3867	465	418	409	306	145	40	29	154	370	551	498	482	(11)
	CDH74	44556	4531	4058	3912	3144	1888	845	794	2177	4747	7426	6104	4930	(12)
	CDH80	22025	2211	1949	1844	1409	633	162	147	866	2635	4397	3338	2434	(13)
(14) Wind	WSAvg	7.1	6.7	6.6	6.8	6.6	6.6	6.5	7.0	7.3	8.0	8.6	7.6	6.9	(14)
(15) Precipitation	PrecAvg	17.8	4.3	3.8	2.3	1.0	0.2	0.0	0.0	0.0	0.2	0.6	1.9	3.1	(15)
	PrecMax	36.4	13.7	14.4	7.6	4.7	1.3	0.7	0.2	0.4	1.3	4.0	4.4	10.3	(16)
	PrecMin	8.7	0.8	0.3	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.1	0.5	(17)
	PrecStd	6.3	2.8	3.3	2.0	1.2	0.3	0.1	0.0	0.1	0.3	0.9	1.3	2.3	(18)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	101.9	98.6	98.7	94.9	91.6	87.1	86.4	93.3	99.3	102.5	103.6	100.1	(19)
		MCWB	67.3	68.0	67.7	67.3	63.7	60.2	57.8	61.8	63.6	65.9	67.0	67.7	(20)
	2%	DB	96.6	95.6	95.4	92.5	88.5	84.4	83.8	90.3	97.0	100.6	100.2	96.9	(21)
		MCWB	70.1	69.6	67.8	66.0	62.7	60.1	57.5	60.4	63.2	64.8	67.1	68.5	(22)
	5%	DB	93.4	93.4	93.0	90.0	86.1	82.0	81.0	88.0	95.2	98.9	97.2	94.4	(23)
		MCWB	70.5	69.7	68.1	65.9	61.7	58.6	56.9	59.5	62.4	64.5	66.8	69.5	(24)
	10%	DB	91.0	91.1	89.7	87.9	83.9	79.1	78.8	85.6	93.2	97.0	94.9	91.5	(25)
		MCWB	70.8	70.2	68.5	66.1	61.4	57.9	56.3	58.6	61.8	64.3	67.2	69.9	(26)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	75.9	75.3	74.7	72.9	68.8	64.3	62.5	66.4	69.3	71.8	73.4	75.1	(27)
		MCDB	87.5	85.5	83.7	85.4	83.9	81.6	80.6	86.9	91.9	94.2	89.4	86.4	(28)
	2%	WB	74.5	74.2	73.5	71.2	66.4	62.4	59.9	63.1	66.7	70.0	71.9	73.7	(29)
		MCDB	86.0	85.3	83.3	83.1	80.7	79.6	79.8	85.0	90.3	91.6	87.6	85.7	(30)
	5%	WB	73.7	73.4	72.6	69.9	64.3	60.4	58.2	61.2	64.9	68.6	70.9	72.8	(31)
		MCDB	84.8	84.9	82.4	81.4	80.6	77.7	78.2	84.1	88.2	89.5	86.4	84.8	(32)
	10%	WB	72.9	72.7	71.8	68.5	62.6	58.8	56.8	59.6	63.5	67.3	70.0	72.0	(33)
		MCDB	83.4	83.6	81.4	80.6	79.9	76.5	76.5	82.6	87.2	88.2	85.3	83.9	(34)
(35) Mean Daily Temperature Range	5% DB	MDBR	19.7	20.8	21.2	25.4	28.8	29.8	30.4	30.5	30.1	26.7	23.7	20.5	(35)
		MCDBR	24.6	25.4	26.5	28.8	30.5	32.3	33.7	32.4	31.8	29.2	27.5	25.1	(36)
	5% WB	MCWBR	7.7	7.9	9.4	11.3	13.5	15.2	16.1	14.5	13.4	11.4	9.8	7.6	(37)
		MCWBR	18.7	19.4	19.2	23.3	27.5	28.7	30.5	30.4	29.7	25.7	22.7	19.8	(38)
(40) Clear-Sky Solar Irradiance	taub		0.383	0.383	0.367	0.336	0.303	0.298	0.303	0.371	0.454	0.447	0.398	0.382	(40)
		taud	2.400	2.407	2.437	2.464	2.500	2.474	2.440	2.243	2.032	2.107	2.303	2.393	(41)
	Ebn at Noon		305	301	298	295	294	290	291	279	268	279	298	305	(42)
		Edh at Noon	41	40	37	33	30	29	31	41	55	53	44	41	(43)
(44) All-Sky Solar Radiation	RadAvg		2141	2043	1921	1797	1654	1487	1562	1789	2015	2176	2136	2109	(44)
	RadStd		168	174	138	119	48	60	46	69	94	129	178	125	(45)

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (0 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page